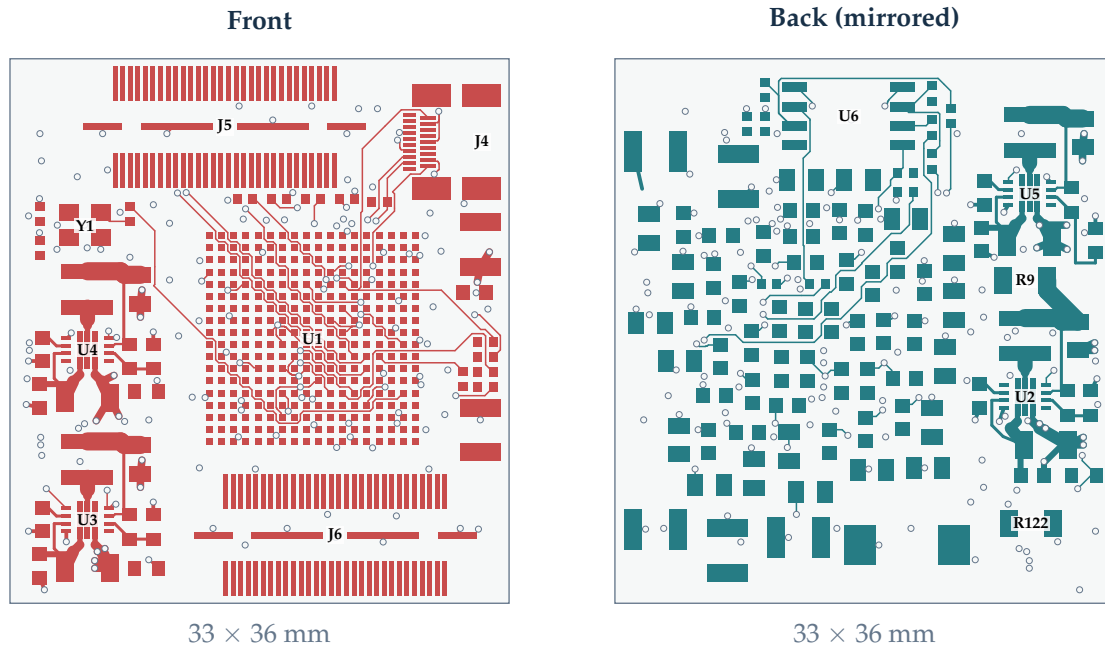


Understanding our 100T FPGA board

The smallest published 33×36 mm design

26 September 2026 / XC7A100T, CSG324, 15×15 mm package



Saved native geometry: front and mirrored back. Outer copper, numbered pad envelopes and vias only; inner layers and ground fills are omitted. This is a review view, not an assembly drawing.

What you will learn. Follow the signal path, understand the seven component groups, read the pin labels, and explain which parts can be reconsidered. Every one of the 130 components and 762 numbered electrical endpoints is indexed in the appendices.

Current correction / limit The board is an incomplete routing checkpoint. U1.L9/L10 must be grounded; four regulator FB2 pins need grounding review. These findings are incorporated here, but the native circuit has not been corrected. There are also 146 assigned-net routing gaps and 331 unassigned endpoints.

How to use this report. Read Sections 1–12 first; use the appendices during questions. The component and pin reference is evidence to consult, not a script to read aloud.

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Two ways to read it

Learn: work through the diagrams and “Explain it aloud” boxes. Terms are defined as they appear.

Present: use the short briefing on page 14; open a detailed component or pin row only when asked. The linked [website](#) supports zooming and searching.

1 Start with the job of the board

The intended system receives signals from eight ASICs (custom recording chips). The routing board brings those signals to this 100T board. FPGA logic is intended to capture data, generate control and timing signals, and format a downstream stream. The XEM8310 is another FPGA module with a USB interface; it is not a conventional microcontroller [9].



Figure 1. Intended system path. This diagram describes the target behavior; full ASIC and cable assignments and end-to-end firmware are unfinished.

Function	Parts	What the group contributes
FPGA logic	1	U1: programmable acquisition and control logic.
Four power rails	39	Four converters and their energy storage, voltage-setting and startup parts.
FPGA supply decoupling	61	Local and bulk supply capacitors around the FPGA power domains.
Boot memory	11	Flash memory and its local supply and signal-support parts.
System clock	4	The 32 MHz oscillator and its supply/edge support.
Configuration and JTAG	11	Startup straps, pull-ups and JTAG support.
External connections	3	J4 powered cable; J5/J6 two 60-contact mezzanines.

Why 130 parts around one chip? The FPGA package contains the digital logic, but it still needs correct power, stable startup states, a configuration image, clocking and physical interfaces. Small supporting components provide those functions.

Read a component reference. U identifies an integrated circuit; C a capacitor; R a resistor; L an inductor; J a connector; Y the oscillator. For example, U1 is the FPGA, while R119 is a resistor in its reference-clock path. These letters are reference labels, not pin numbers.

Explain it aloud This is the interface between the recording-chip routing board and the downstream FPGA module. The surrounding parts make power, startup, timing and connection possible.

2 Power: one cable input, four local supplies

A **rail** is a named power-supply connection. The FPGA core, auxiliary circuits and I/O banks use different rails. A **bank** is a group of FPGA I/O pins sharing a supply and electrical constraints; it is not an independently programmable voltage source.

The current architecture proposes nominal 12 V at J4.19. Each TPS62135 converter and inductor steps that voltage down. The 1.5 V ASIC-facing and 2.5 V link-bank choices still require interface validation; they are not universal requirements of every Artix-7 application [1,2].

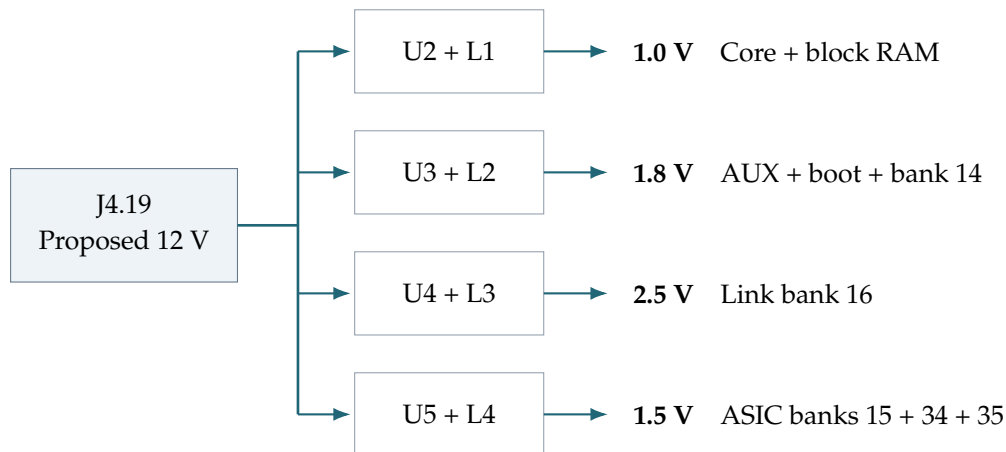


Figure 2. Supply destinations. This is a functional diagram, not a routed power or enable diagram. R9 separates U2’s local output from the core rail; R122 does the same for U3’s distributed 1.8 V rail.

What each converter needs. Its inductor transfers energy; input/output capacitors support the switching-current loops; a feedback divider sets the output; the soft-start network controls the startup transition. Placement and current-return paths matter as much as logical connectivity [1].

Startup order. The present enable chain is PG_CORE → U3, then PG_AUX → U4/U5. These two sequencing nets are connected in the saved checks. Startup and shutdown waveforms still need measurement. R12’s PGOOD_IO signal has no consumer.

Why the local names differ. U2’s local sense node is VCORE_REG_1V025; the distributed rail after R9 is VCCINT_1V0. R9 and R122 are 15 milliohm isolation resistors, not zero-ohm jumpers. For illustration only, 1 A through 15 milliohm drops 15 mV; the actual load and copper loss need analysis. Do not bypass either resistor during routing.

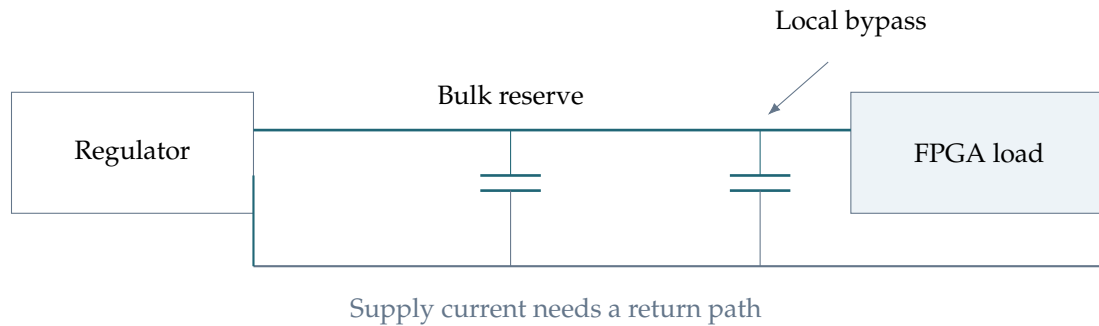
Current correction / limit The headboard does not yet have a completed input-protection solution. A powered cable and carrier must be designed and qualified; an unmodified XEM8310 is not established as this board’s power source.

Explain it aloud The cable voltage is converted into the rails required by our chosen core and interface architecture. Connected power-good wires do not yet prove correct power-up at the FPGA pins.

3 Capacitors: why so many small parts?

A capacitor stores charge. Close to a switching load, it can supply a brief current pulse while the regulator and wider power network respond. **Decoupling** means providing that local supply support and limiting unwanted coupling through supply impedance.

There are **83 capacitors in the saved board**: 61 belong to the FPGA supply-decoupling group, and 22 support converters, cable entry, flash or oscillator circuits. The figure below illustrates a function, not a literal trace path.



FPGA domain	Capacitors	Count
Core	C20, C22-C35	15
Block RAM	C21, C36, C37	3
Auxiliary	C40-C46	7
Configuration bank 0	C50	1
Bank 14	C51-C56, C92	7
Bank 16	C57-C63	7
Banks 15 / 34 / 35	C64-C81	18
Shared ASIC-bank bulk	C82	1
XADC-related supply	C90, C91	2

Why different values and locations? The entire mounted network has capacitance, resistance and inductance. Larger nominal capacitance is not automatically better at every frequency, and a long connection can reduce the value of a nearby capacitor. Two parts on the same net can serve different current paths [3].

Can we remove some? The required function is an adequate power-distribution network. The exact count is not proven minimal. Reduction needs an application load model, effective capacitance, impedance/placement analysis and suitable measurements. C92 supports bank 14 after the voltage-domain split; removing it simply because other 1.8 V capacitors exist is not justified. C21's recorded part choice also needs an orderable replacement review.

Explain it aloud These are local energy reserves, not decorative extras. We can optimize the network, but we need evidence that the FPGA supply remains within limits after the change.

4 Startup, memory, clock and programming

Configuration is loading the FPGA's logic design. It is separate from the application data captured later. U6 is the 1.8 V MX25U12835F flash that stores a configuration image for independent startup [4,5]. Removing it requires another configuration source at every power-up.

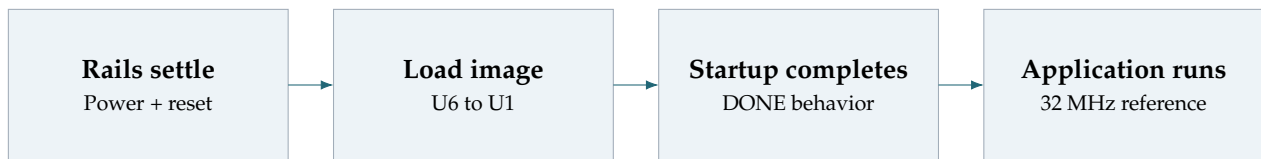


Figure 3. Conceptual startup sequence, not measured timing. The oscillator can run before the application; it is not switched on by DONE in this circuit.

Parts	Purpose	What to remember
U6; C100/C101	Boot storage and local supply support	A stored image and verified flash/configuration settings are still needed.
R100-R107	Flash startup levels and signal paths	U6.7 is RESET/SIO3 for this exact part. Zero-ohm links still carry a connection.
Y1; C102/C103; R119	1.8 V oscillator, bypass and clock edge	32 MHz reference; damping value needs signal-integrity review.
R108 / R109 R110	PROGRAM_B / INIT_B pull-ups Additional DONE pull-up	Keep the specified external pull-up functions. Conditional: DONE has an internal pull-up; review loading/startup before removal.
R111-R114	SPI mode and PUDC_B startup straps	Keep the required states; reviewed copper ties could replace some resistor footprints.
R115-R118; J4	JTAG bias and TDO path	Dedicated FPGA programming pins share the custom cable connector.

Read one actual signal path. Y1.3 and R119.1 share OSC_32M_RAW. R119.2 and U1.P17 share CLK_32MHZ. The resistor deliberately separates two net names. The signal nets are connected in the saved native check; rail/ground completion and waveform quality remain open.

A pull-up is not a wire replacement. It biases a signal high while allowing a device to drive it low. A fixed mode strap has a different role. That is why all startup resistors cannot be removed using one general rule [4].

Explain it aloud Flash stores the design, the oscillator supplies a time reference, and startup/JTAG parts make configuration predictable and accessible. No cold boot or programming test has yet been demonstrated on this board.

5 The 117 signals and three connectors

The corrected ASIC interface budget is **117 logical signals**. The 88-output group consists of eight sets of 11: each set includes eight data outputs, a returned clock, READ and SYNC. It is not 88 independent sampled data bits. Gerald's tutorial establishes useful framing, but electrical limits and several timing details remain unresolved [8].

Signal group	Count	Interpretation
ASIC output/interface lines	88	8 ASICs x 11; includes data and associated timing/handshake.
SPI L/R streams	12	Stimulation and non-stimulation configuration data streams.
Shared controls	9	CHIP_RESET, AC_IN, IMP_TST, FE_RESET, SPI_LATCH, STIM_CLK, STIM_START, STIM_EN, STIM_CHB.
Board clocks	4	One recording-board clock per board.
Additional SPI clocks	4	Correction to the original 113-signal count.
Total	117	Excludes grounds, power and shielding.

J5 and J6: routing-board connection. Two 60-contact mezzanines provide 120 numbered signal contacts, plus separate ground-blade lands. The allocation proposal is 117 signals plus three reserved contacts. All 120 remain unassigned in the native design, so those three are not yet implemented permanent spares.

J4: powered cable and JTAG. This micro-HDMI-style connector has a custom electrical assignment. It is incompatible with ordinary HDMI equipment. The eight lane contacts are reserved for a proposed data link; they are still unassigned in CAD.

J4 contacts	Saved assignment	Meaning
19	LINK_12V	Proposed nominal 12 V input.
1	VCCAUX_1V8	1.8 V target-voltage reference for programming.
2 / 15 / 17 / 18	TMS / TDI / TCK / TDO	Dedicated target JTAG connections.
16; 4 / 7 / 10 / 13; SH	GND	DC return, pair shields and shell.
3/5; 6/8; 9/11; 12/14	Unassigned	Four proposed differential-pair positions.

Current correction / limit The XEM8310 carrier contacts, cable continuity, voltage/current limits, link timing, termination and target-JTAG bridge remain to be designed or qualified. AC_IN and IMP_TST do not yet have established pad functions and safe electrical limits.

Explain it aloud J5 and J6 carry our ASIC interface; J4 combines power and programming with reserved data contacts. None of these three connectors is an unexplained extra header.

6 Unused function does not mean leave it open

A **pin** is an electrical terminal; a **pad** is its physical copper landing. Some ground and shell terminals have multiple pads. A **net** names terminals intended to be electrically joined. A **via** connects copper between board layers.

The current 762 numbered endpoints comprise 425 assigned endpoints, 331 unassigned endpoints and six marked NC (no-connect) in the saved CAD. This report adds the electrical review overlay below. It does not silently turn unfinished interfaces into approved no-connects.

Code	Read it as	What it establishes
C	Assigned; whole net connected	Native connectivity only, not waveform quality or working hardware.
P	Assigned; net has gaps	At least one gap exists somewhere on this net. This individual pad may already be connected locally.
U	Awaiting assignment	A final signal/net has not been implemented. Do not call every U pin a permanent spare.
G	Ground required; saved NC	U1.L9 DXN_0 and U1.L10 DXP_0 require GND per AMD UG475.
R	Grounding review; saved NC	U2-U5 pin 4 FB2: unused switched-divider outputs. TI single-divider example grounds them.

Current correction / limit **U1.L9 and U1.L10 need correction.** The temperature-diode function is unused, but AMD's package guidance requires the pins tied to GND [7]. Their current NC markers are incorrect final wiring. The source PCB remains unchanged in this report.

FB2 is a different case. U2.4, U3.4, U4.4 and U5.4 are optional switched-feedback outputs. No divider branch is attached here. TI's Figure 5 grounds FB2 in its single-divider circuit; this audit therefore marks a grounding review, without pretending the current NC treatment has explicit manufacturer approval [1].

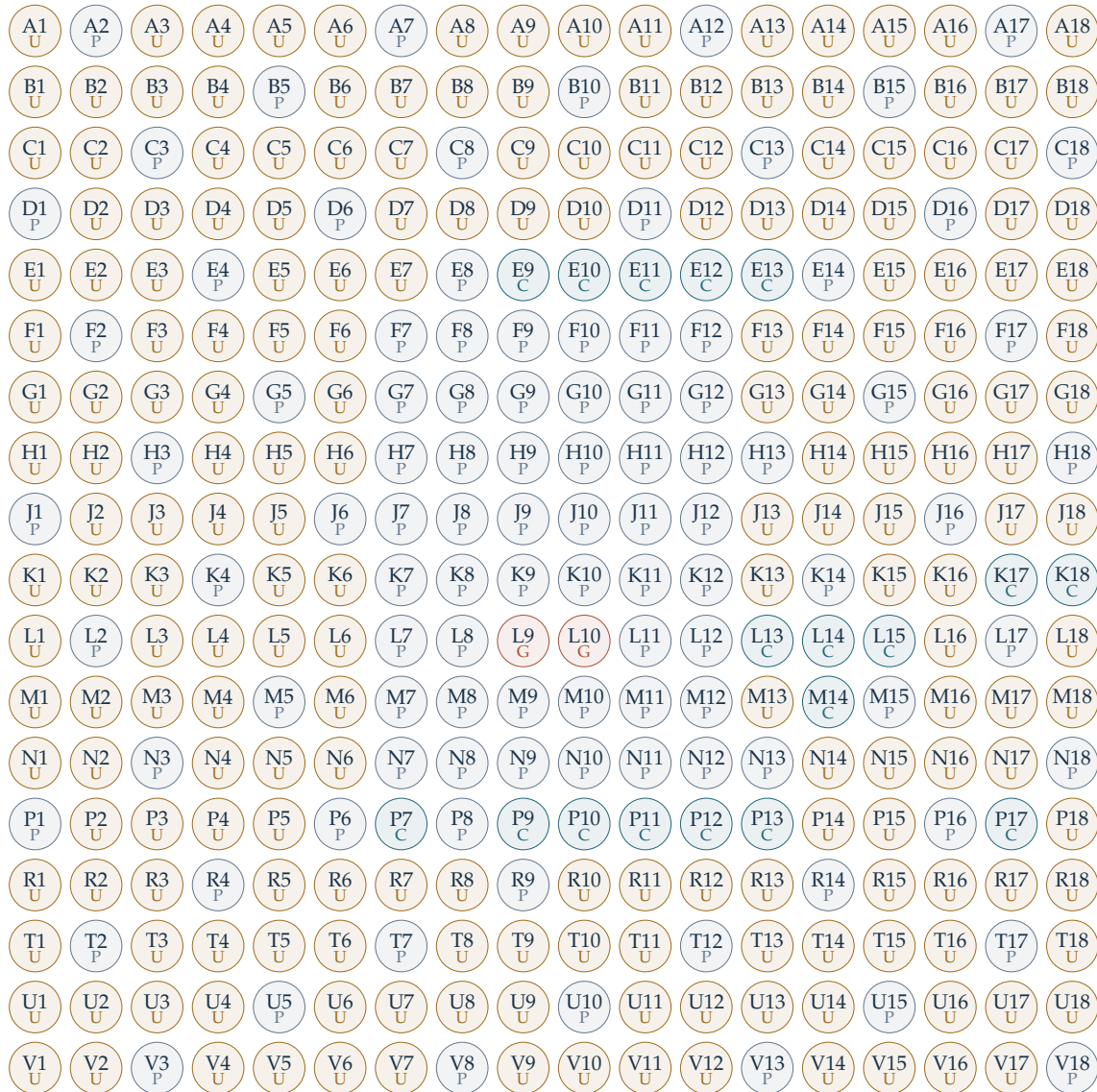
331 unassigned endpoints are separate from 146 routing gaps. The unassigned set is 203 U1 user-I/O balls + 120 J5/J6 contacts + eight J4 lane contacts. A signal with no assignment cannot be counted as a missing connection on an already assigned net.

Do not double-count physical lands. The 762 electrical endpoint IDs represent 771 numbered pad records. Another 40 unnumbered pad records bring the physical total to 811. The 130 components are neither 130 pins nor 130 nets.

Explain it aloud I use one label for unassigned signals, another for incomplete copper, and explicit correction labels for unused functions that still need a ground connection or review.

7 Read the FPGA ball map

Every one of U1's 324 balls is shown below using native pad coordinates, viewed from the front of the PCB. A1 is at the top left in this saved layout. This is not a bottom-of-package or connector mating-face drawing. Each circle shows its ball name and the status code defined on page 8.



U: unassigned **P:** assigned net has gaps **C:** assigned net connected **G:** ground required

Try three examples. U1.A1 is awaiting assignment. U1.P17 is the assigned 32 MHz clock input. U1.L9/L10 are the two red ground corrections. The complete function, bank and actual net for every ball are listed in Appendix B.

203 U labels do not establish the spare count. The 117 ASIC assignments and the proposed downstream link still need to be placed into legal balls, banks and clock resources, then checked in the implementation tools.

8 Read the connector pad maps

These maps preserve the actual relative pad centers in the current PCB. Contacts are identified individually. Colors use the same U/P/C legend as the FPGA map. The drawings are for locating pins in the PCB top view, **not for approving a cable or mating-board pin order.**

J5 – first 60-contact mezzanine



J6 – second 60-contact mezzanine



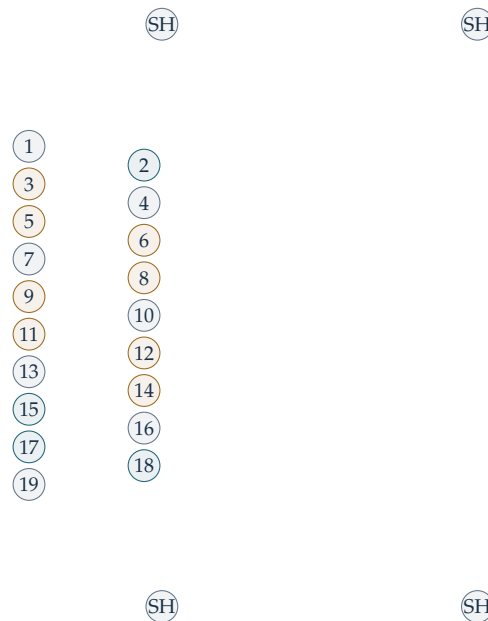
All 120 numbered J5/J6 contacts are amber: unassigned. Each set of G lands shares one electrical G endpoint. The shared GND net still has gaps.

Before a mating board can be released: reconcile connector top/bottom views, connector orientation and the final contact-to-FPGA map on both sides. A clear drawing does not by itself establish that the two boards will mate electrically.

How to read the rows. The saved orientation places even-numbered contacts in the upper row and odd-numbered contacts in the lower row. The layout of the mating connector may be viewed from the opposite side; confirm the manufacturer's drawing rather than copying this picture as a mating-face map.

9 J4: read the custom cable contact map

J4 uses the shape of a micro-HDMI connector with a custom electrical assignment. View the map from the PCB front, as for J5/J6. Repeated SH labels are shell pads on one electrical endpoint. Amber contacts are unassigned, gray contacts belong to nets with gaps, and teal contacts are on fully connected assigned nets.



The saved assignments, one contact at a time

Endpoint	Saved net / state	Code
J4.1	VCCAUX_1V8	P
J4.2	JTAG_TMS	C
J4.3	Unassigned	U
J4.4	GND	P
J4.5	Unassigned	U
J4.6	Unassigned	U
J4.7	GND	P
J4.8	Unassigned	U
J4.9	Unassigned	U
J4.10	GND	P
J4.11	Unassigned	U
J4.12	Unassigned	U
J4.13	GND	P
J4.14	Unassigned	U
J4.15	JTAG_TDI	C
J4.16	GND	P
J4.17	JTAG_TCK	C
J4.18	JTAG_TDO	C
J4.19	LINK_12V	P
J4.SH	GND	P

Read an example. J4.17 is JTAG TCK; J4.19 is the proposed 12 V input. J4.3 and J4.5 are an unassigned lane-pair position. The eight lane contacts are not connected to FPGA balls yet. Ground returns, cable orientation, carrier mapping and electrical limits still need verification. Do not connect standard HDMI equipment.

10 Which components can we reconsider?

The review asks two separate questions: **is the function needed?** and **is this exact implementation the smallest adequate one?** A part can support a necessary function while its value, quantity or footprint remains open to optimization.

Decision class	Parts	Interpretation
Keep function	32	Essential in this topology; an equivalent redesign is possible.
Depends on system choice	13	Needed by the chosen boot, clock, rail or connection strategy.
Check before reducing	84	Supports power, startup or signal quality; exact individual minimum is unproven.
Removal candidate	1	R12: its present status signal has no functional consumer.

Candidate / group	Reason	Required action before claiming success
R12	PGOOD_IO has no consumer.	Remove/review its branch deliberately, or give it a monitoring function.
R110	DONE already has an internal pull-up.	Check startup configuration and loading before removing the external pull-up.
R111-R114	Mode/PUDC_B straps can use fixed ties.	Preserve the intended logic states in the redesigned copper.
R104 / R106 / R107	Fitted zero-ohm signal links.	Replace by reviewed copper; an empty footprint would break the signal.
Capacitors and damping parts	Network performance matters more than count.	Demonstrate supply/edge behavior with the actual load, routing and assembly.
J4 / J5 / J6	Each serves the chosen system interface.	No separate J2/J3 debug header or test-point component remains to remove.

Why not shrink the outline immediately? The 15×15 mm FPGA is only one object in a 33×36 mm board. Remaining space may be required for BGA escape, rail distribution, ground returns, connector access and assembly tolerance. The present minimum courtyard gap of about 0.010 mm is not an assembler-approved clearance. J4's drawn body adds about 0.65 mm beyond one edge.

The board is the **smallest published checkpoint**, not a demonstrated global minimum or a qualified complete assembly. There is no proof that reducing the count by one part reduces the outline.

Explain it aloud We have one clear removal candidate and several conditional simplifications. We should optimize the circuit and routing against measurable requirements, not infer that every empty patch of the drawing is wasted area.

11 What has been checked, and what has not?

DRC checks PCB geometry against the saved design rules. **ERC** checks schematic electrical-rule categories. **Parity** checks correspondence between the saved schematic and PCB. These are useful checks, but a wrong connection can be copied consistently into both files.

Check	Saved result	Meaning / limitation
Components / endpoints	130 / 762	Explained and cross-checked against saved evidence.
Physical DRC / parity	0 / 0	No reported geometry/parity findings under the saved rules; not electrical approval.
Assigned-net connectivity	146 gaps	Power/ground routing remains; this count excludes unassigned interfaces.
Unassigned endpoints	331	203 FPGA + 120 mezzanine + eight cable contacts.
Schematic ERC	331 open pins + 5 warnings	Known open interfaces and symbol-type warnings remain.
Boot / clock / JTAG nets	25 nets connected	Copper connectivity; no demonstrated programming or cold boot.
Power-good nets	2 sequencing + 1 status	The status net is unconsumed; it is not a third enable stage.
Built hardware	Not tested	No rail, thermal, capture or sustained-link measurements from this board.

Where the 146 gaps are recorded

Net	Gaps	Net	Gaps
GND	8	LINK_12V	9
VAUX_REG_1V803	1	VCCAUX_1V8	49
VCCINT_1V0	34	VCC_ASIC_1V5	37
VCC_LINK_2V5	8	Total	146

U4.11 is a useful example. Its schematic net is GND, but its VSEL copper strap is still open. Pin 3 is the converter's high-current GND terminal; the two pin functions should not be confused. The U1 diode-pin error is different again: those pins need a corrected net assignment, so it is not included in the 146 assigned-net count.

Explain it aloud The saved layout passes its geometry and parity checks. That does not make the circuitry correct or complete: we still have power routing, interface assignments, electrical corrections and hardware tests ahead.

12 A briefing you can give to your professor

Explain it aloud Our smallest published PCB is 33 by 36 millimeters and uses a 100T FPGA in a 15 by 15 millimeter package. It is intended to capture the eight-ASIC interface and send data to the XEM8310 receiver. Its 130 components fall into seven functions: FPGA logic, power, decoupling, boot memory, clock, startup/programming and three connectors.

Continue with the design choices.

“The four local supplies support the core, auxiliary/configuration circuits and two interface-voltage domains. J5 and J6 are the requested two 60-contact mezzanines. J4 combines custom cable power and JTAG, with data contacts still to assign. There are no separate debug headers, user LEDs or standalone test points in this version.”

Then state the honest result.

“The component audit found that R12 has no current consumer. Other reductions require specific circuit or performance checks. We also found two FPGA pins that must be grounded and four regulator pins needing grounding review. These findings are documented, but the native circuit is unchanged. The design still has 146 assigned-net gaps and 331 unassigned endpoints.”

Questions to be ready for

Question	Answer
Is every capacitor mandatory?	The network function is necessary. We have not demonstrated that every individual capacitor is the minimum adequate population.
Why use flash?	It enables autonomous configuration after power-up. Removing it requires another configuration source.
Are 203 FPGA pins spare?	No. They are unassigned today; the application and link maps have not been implemented.
Does zero DRC mean it works?	No. It checks saved geometry rules. Electrical requirements and powered behavior are separate.
Can 120 contacts carry 117 signals?	The count fits with three proposed reserves plus separate ground lands. Mapping, electrical and timing validation remain.
What is the next evidence needed?	Approved ASIC pad/timing data; closed pin/cable contracts; corrected schematic; complete qualified routing; then prototype measurements.

Use the report during questions. Find a part in Appendix A; find a ball or contact in Appendices B–D. A P label means the net has a gap somewhere, while G/R explicitly identifies a pending connection correction or review.

13 Sources, revision and reading limits

This is a learning report derived from the current saved circuit and the later component/unused-pin audit. It supersedes the earlier report's interpretation of the six NC markers; it does not modify native CAD or claim fabrication readiness. Manufacturer statements and the actual saved net assignments are kept separate from engineering choices that still need validation.

Native board SHA-256

ae601df3925e5ec5e90c8a2e747356e432d0558e820199cf7c01c6bf8a7fef96

Source records. The report uses the current component list, all-pin CSV, native readback, routing snapshot, component necessity audit and pin-status dataset. The portable source archive includes these ledgers and a coverage record. All 130 components and 762 endpoint IDs are covered; the appendices contain each endpoint once in their primary tables.

Primary references and project evidence

- [1] [Texas Instruments, TPS62135/TPS621351 datasheet](#). Pin table p3; single-divider circuit Figure 5 p16; distributed capacitance pp32–33; layout p36. FB2 grounding remains a review finding, not an implemented correction.
- [2] [AMD DS181, Artix-7 DC and AC characteristics](#). Supply limits, sequencing and ramp requirements.
- [3] [AMD UG483, 7 Series PCB Design Guide](#). Capacitor guidance, placement and power-network analysis.
- [4] [AMD UG470, 7 Series Configuration User Guide](#). Configuration voltages, mode/PUDC_B straps, PROGRAM_B/INIT_B pull-ups and DONE behavior.
- [5] [Macronix MX25U12835F, 1.8 V flash, v1.9](#). Exact-part pin 7 is RESET/SIO3.
- [6] [Abracon ASE3 series oscillator datasheet](#). Supply and four-pin mapping. A generic bypass note has mismatched 7/14-pin wording; the selected four-pin table is used here.
- [7] [AMD UG475 v1.20, Table 1-12, p26](#). Unused DXP/DXN pins require GND. This is the basis for the two explicit corrections.
- [8] [Gerald's ASIC interface slides, pp14–17 and 25–28](#). Recording/SPI tutorial behavior. Timing/channel-label conflicts and missing electrical limits are preserved as open questions. The team's later correction adds four SPI clocks to reach 117 signals.
- [9] [Opal Kelly XEM8310 documentation](#). Downstream FPGA module and interface context; no automatic target-power or JTAG bridge is assumed.
- [10] [Controlpaths CP SOM ONE, pinned reference revision](#). Power/schematic/layout reference; its device, package and load differ from this board.

Status boundary. No new ERC/DRC or powered hardware test was performed for this document. The numerical CAD results refer to the frozen source board. The later manufacturer review adds the diode-ground and FB2 findings. Routing, ASIC/XEM8310 electrical contracts, firmware, timing, power integrity, thermal and manufacturing approval remain unfinished.

A Every component: purpose and removal consequence

Use this as a lookup table. The decision categories apply to the current topology, not to every possible FPGA design. K = keep function; S = system choice; Q = qualify before reduction; X = removal candidate. Values are saved design values, not a released procurement list.

Every two-terminal part lists both actual endpoints. Larger devices point to the complete tables that follow. C/P/U/G/R are pin/net status codes from page 8; G/R replace any implication that a saved NC marker is approved final wiring.

Part / value	Role; consequence if removed	Actual endpoint: net [status]
C1 [Q] 47u 25V X5R	Cable-entry bulk reservoir Reduces cable-entry energy reserve; depends on cable impedance, inrush and source behavior.	C1.1: LINK_12V [P] C1.2: GND [P]
C2 [Q] 100n 100V X7R	Cable-entry high-frequency bypass Changes entry-node high-frequency impedance.	C2.1: LINK_12V [P] C2.2: GND [P]
C3 [K] 22u 25V X5R	Converter input reservoir The local current loop relies on distant capacitors and cable inductance; input ripple and unstable startup become possible.	C3.1: LINK_12V [P] C3.2: GND [P]
C4 [Q] 100n 100V X7R	High-frequency input bypass Changes high-frequency input impedance. The larger capacitor may not behave identically at fast edges.	C4.1: LINK_12V [P] C4.2: GND [P]
C5 [K] 22u 16V X5R	Converter local output capacitor The output filter and sensed node lose their intended local capacitor; supply ripple and control behavior change.	C5.1: VCORE_REG_1V025 [C] C5.2: GND [P]
C6 [Q] 10n 50V X7R	Soft-start timing capacitor The converter uses its fast internal startup behavior; inrush and FPGA rail ramp/sequence change.	C6.1: SS_CORE [C] C6.2: GND [P]
C7 [K] 22u 25V X5R	Converter input reservoir The local current loop relies on distant capacitors and cable inductance; input ripple and unstable startup become possible.	C7.1: LINK_12V [P] C7.2: GND [P]
C8 [Q] 100n 100V X7R	High-frequency input bypass Changes high-frequency input impedance. The larger capacitor may not behave identically at fast edges.	C8.1: LINK_12V [P] C8.2: GND [P]
C9 [K] 22u 16V X5R	Converter local output capacitor The output filter and sensed node lose their intended local capacitor; supply ripple and control behavior change.	C9.1: VAUX_REG_1V803 [P] C9.2: GND [P]
C10 [Q] 10n 50V X7R	Soft-start timing capacitor The converter uses its fast internal startup behavior; inrush and FPGA rail ramp/sequence change.	C10.1: SS_AUX [C] C10.2: GND [P]
C11 [K] 22u 25V X5R	Converter input reservoir The local current loop relies on distant capacitors and cable inductance; input ripple and unstable startup become possible.	C11.1: LINK_12V [P] C11.2: GND [P]
C12 [Q] 100n 100V X7R	High-frequency input bypass Changes high-frequency input impedance. The larger capacitor may not behave identically at fast edges.	C12.1: LINK_12V [P] C12.2: GND [P]
C13 [K] 22u 16V X5R	Converter local output capacitor The output filter and sensed node lose their intended local capacitor; supply ripple and control behavior change.	C13.1: VCC_LINK_2V5 [P] C13.2: GND [P]
C14 [Q] 10n 50V X7R	Soft-start timing capacitor The converter uses its fast internal startup behavior; inrush and FPGA rail ramp/sequence change.	C14.1: SS_CFG [C] C14.2: GND [P]
C15 [K] 22u 25V X5R	Converter input reservoir The local current loop relies on distant capacitors and cable inductance; input ripple and unstable startup become possible.	C15.1: LINK_12V [P] C15.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
C16 [Q] 100n 100V X7R	High-frequency input bypass Changes high-frequency input impedance. The larger capacitor may not behave identically at fast edges.	C16.1: LINK_12V [P] C16.2: GND [P]
C17 [K] 22u 16V X5R	Converter local output capacitor The output filter and sensed node lose their intended local capacitor; supply ripple and control behavior change.	C17.1: VCC_ASIC_1V5 [P] C17.2: GND [P]
C18 [Q] 10n 50V X7R	Soft-start timing capacitor The converter uses its fast internal startup behavior; inrush and FPGA rail ramp/sequence change.	C18.1: SS_ASIC [C] C18.2: GND [P]
C20 [Q] 330u 2.5V polymer ESR25m	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C20.1: VCCINT_1V0 [P] C20.2: GND [P]
C21 [Q] 100u 6.3V X5R	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C21.1: VCCINT_1V0 [P] C21.2: GND [P]
C22 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C22.1: VCCINT_1V0 [P] C22.2: GND [P]
C23 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C23.1: VCCINT_1V0 [P] C23.2: GND [P]
C24 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C24.1: VCCINT_1V0 [P] C24.2: GND [P]
C25 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C25.1: VCCINT_1V0 [P] C25.2: GND [P]
C26 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C26.1: VCCINT_1V0 [P] C26.2: GND [P]
C27 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C27.1: VCCINT_1V0 [P] C27.2: GND [P]
C28 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C28.1: VCCINT_1V0 [P] C28.2: GND [P]
C29 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C29.1: VCCINT_1V0 [P] C29.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
C30 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C30.1: VCCINT_1V0 [P] C30.2: GND [P]
C31 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C31.1: VCCINT_1V0 [P] C31.2: GND [P]
C32 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C32.1: VCCINT_1V0 [P] C32.2: GND [P]
C33 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C33.1: VCCINT_1V0 [P] C33.2: GND [P]
C34 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C34.1: VCCINT_1V0 [P] C34.2: GND [P]
C35 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C35.1: VCCINT_1V0 [P] C35.2: GND [P]
C36 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C36.1: VCCINT_1V0 [P] C36.2: GND [P]
C37 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C37.1: VCCINT_1V0 [P] C37.2: GND [P]
C40 [Q] 47u 6.3V X7R	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C40.1: VCCAUX_1V8 [P] C40.2: GND [P]
C41 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C41.1: VCCAUX_1V8 [P] C41.2: GND [P]
C42 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C42.1: VCCAUX_1V8 [P] C42.2: GND [P]
C43 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C43.1: VCCAUX_1V8 [P] C43.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
C44 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C44.1: VCCAUX_1V8 [P] C44.2: GND [P]
C45 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C45.1: VCCAUX_1V8 [P] C45.2: GND [P]
C46 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C46.1: VCCAUX_1V8 [P] C46.2: GND [P]
C50 [Q] 47u 6.3V X7R	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C50.1: VCCAUX_1V8 [P] C50.2: GND [P]
C51 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C51.1: VCCAUX_1V8 [P] C51.2: GND [P]
C52 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C52.1: VCCAUX_1V8 [P] C52.2: GND [P]
C53 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C53.1: VCCAUX_1V8 [P] C53.2: GND [P]
C54 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C54.1: VCCAUX_1V8 [P] C54.2: GND [P]
C55 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C55.1: VCCAUX_1V8 [P] C55.2: GND [P]
C56 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C56.1: VCCAUX_1V8 [P] C56.2: GND [P]
C57 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C57.1: VCC_LINK_2V5 [P] C57.2: GND [P]
C58 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C58.1: VCC_LINK_2V5 [P] C58.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
C59 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C59.1: VCC_LINK_2V5 [P] C59.2: GND [P]
C60 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C60.1: VCC_LINK_2V5 [P] C60.2: GND [P]
C61 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C61.1: VCC_LINK_2V5 [P] C61.2: GND [P]
C62 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C62.1: VCC_LINK_2V5 [P] C62.2: GND [P]
C63 [Q] 47u 6.3V X7R	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C63.1: VCC_LINK_2V5 [P] C63.2: GND [P]
C64 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C64.1: VCC_ASIC_1V5 [P] C64.2: GND [P]
C65 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C65.1: VCC_ASIC_1V5 [P] C65.2: GND [P]
C66 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C66.1: VCC_ASIC_1V5 [P] C66.2: GND [P]
C67 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C67.1: VCC_ASIC_1V5 [P] C67.2: GND [P]
C68 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C68.1: VCC_ASIC_1V5 [P] C68.2: GND [P]
C69 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C69.1: VCC_ASIC_1V5 [P] C69.2: GND [P]
C70 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C70.1: VCC_ASIC_1V5 [P] C70.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
C71 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C71.1: VCC_ASIC_1V5 [P] C71.2: GND [P]
C72 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C72.1: VCC_ASIC_1V5 [P] C72.2: GND [P]
C73 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C73.1: VCC_ASIC_1V5 [P] C73.2: GND [P]
C74 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C74.1: VCC_ASIC_1V5 [P] C74.2: GND [P]
C75 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C75.1: VCC_ASIC_1V5 [P] C75.2: GND [P]
C76 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C76.1: VCC_ASIC_1V5 [P] C76.2: GND [P]
C77 [Q] 4.7u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C77.1: VCC_ASIC_1V5 [P] C77.2: GND [P]
C78 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C78.1: VCC_ASIC_1V5 [P] C78.2: GND [P]
C79 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C79.1: VCC_ASIC_1V5 [P] C79.2: GND [P]
C80 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C80.1: VCC_ASIC_1V5 [P] C80.2: GND [P]
C81 [Q] 470n 6.3V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C81.1: VCC_ASIC_1V5 [P] C81.2: GND [P]
C82 [Q] 47u 6.3V X7R	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C82.1: VCC_ASIC_1V5 [P] C82.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
C90 [Q] 100n 16V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C90.1: VCCAUX_1V8 [P] C90.2: GND [P]
C91 [Q] 1u 10V X7R	Local supply bypass Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C91.1: VCCAUX_1V8 [P] C91.2: GND [P]
C92 [Q] 47u 6.3V X7R	Bulk energy reserve Less local stored charge and changed power-network impedance; may cause supply noise or transient failures. Removing one is not automatically safe or automatically fatal.	C92.1: VCCAUX_1V8 [P] C92.2: GND [P]
C100 [K] 100n	Flash local supply bypass Removes the smallest local flash reservoir; remote rail capacitance does not guarantee an equivalent short current loop.	C100.1: VCCAUX_1V8 [P] C100.2: GND [P]
C101 [Q] 1u	Flash local bulk reservoir Less local reserve; flash current transients must then be met by C100 and the rail network.	C101.1: VCCAUX_1V8 [P] C101.2: GND [P]
C102 [K] 10n	Oscillator local supply bypass Removes its intended shortest bypass path.	C102.1: VCCAUX_1V8 [P] C102.2: GND [P]
C103 [Q] 100n	Additional oscillator bypass Less local supply filtering; consequence depends on supply impedance and clock sensitivity.	C103.1: VCCAUX_1V8 [P] C103.2: GND [P]
J4 [S] CUSTOM POWER + JTAG / DATA TBD	Shared power / JTAG / future data cable The selected power-entry and external programming access disappear.	Complete pins: Appendix C.
J5 [S] QSH-030-01-L-D-A	60-contact ASIC mezzanine The proposed 117-contact interface no longer fits in the remaining 60 contacts.	Complete pins: Appendix C.
J6 [S] QSH-030-01-L-D-A	60-contact ASIC mezzanine The proposed 117-contact interface no longer fits in the remaining 60 contacts.	Complete pins: Appendix C.
L1 [K] 1uH 5.4A Isat30%	Buck energy-storage inductor Open circuit cuts power to the rail; replacing with a short exposes the load to switching pulses.	L1.1: SW_CORE [C] L1.2: VCORE_REG_1V025 [C]
L2 [K] 1uH 5.4A Isat30%	Buck energy-storage inductor Open circuit cuts power to the rail; replacing with a short exposes the load to switching pulses.	L2.1: SW_AUX [C] L2.2: VAUX_REG_1V803 [P]
L3 [K] 1uH 5.4A Isat30%	Buck energy-storage inductor Open circuit cuts power to the rail; replacing with a short exposes the load to switching pulses.	L3.1: SW_CFG [C] L3.2: VCC_LINK_2V5 [P]
L4 [K] 1uH 5.4A Isat30%	Buck energy-storage inductor Open circuit cuts power to the rail; replacing with a short exposes the load to switching pulses.	L4.1: SW_ASIC [C] L4.2: VCC_ASIC_1V5 [P]
R1 [K] 32.4k 0.1%	Core / block ram feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R1.1: VCORE_REG_1V025 [C] R1.2: FB_CORE [C]
R2 [K] 69.8k 0.1%	Core / block ram feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R2.1: FB_CORE [C] R2.2: GND [P]
R3 [K] 110k 0.1%	Auxiliary / configuration feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R3.1: VAUX_REG_1V803 [P] R3.2: FB_AUX [C]
R4 [K] 69.8k 0.1%	Auxiliary / configuration feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R4.1: FB_AUX [C] R4.2: GND [P]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
R5 [K] 180k 0.1%	2.5 v link bank feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R5.1: VCC_LINK_2V5 [P] R5.2: FB_CFG [C]
R6 [K] 69.8k 0.1%	2.5 v link bank feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R6.1: FB_CFG [C] R6.2: GND [P]
R7 [K] 80.6k 0.1%	1.5 v asic-facing banks feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R7.1: VCC_ASIC_1V5 [P] R7.2: FB_ASIC [C]
R8 [K] 69.8k 0.1%	1.5 v asic-facing banks feedback divider The intended feedback ratio is lost; the rail may be wrong, too low or driven high.	R8.1: FB_ASIC [C] R8.2: GND [P]
R9 [K] 15m 1% 1W	Distributed-capacitance isolation Removing it opens the rail; shorting it defeats the deliberately separated local and distributed capacitor networks.	R9.1: VCORE_REG_1V025 [C] R9.2: VCCINT_1V0 [P]
R10 [K] 100k	Power-up sequencing pull-up The downstream enable no longer receives its designed high level.	R10.1: LINK_12V [P] R10.2: PG_CORE [C]
R11 [K] 100k	Power-up sequencing pull-up The downstream enable no longer receives its designed high level.	R11.1: LINK_12V [P] R11.2: PG_AUX [C]
R12 [X] 100k	Unconsumed I/O-good status pull-up No existing rail-enable or data path is removed. The status branch would lose its defined high state.	R12.1: VCCAUX_1V8 [P] R12.2: PGOOD_IO [C]
R100 [K] 2.4k	Flash chip-select pull-up Flash chip select loses its external defined idle-high state during startup.	R100.1: VCCAUX_1V8 [P] R100.2: FLASH_CS_B [C]
R101 [S] 4.7k	Flash WP / DQ2 default high The flash write-protect/data pin may lose its defined default level when FPGA outputs are undriven.	R101.1: VCCAUX_1V8 [P] R101.2: FLASH_DQ2 [C]
R102 [S] 4.7k	Flash RESET / DQ3 default high The reset/data pin may lose its defined default level before configuration.	R102.1: VCCAUX_1V8 [P] R102.2: FLASH_DQ3 [C]
R103 [Q] 33	Flash clock source damping If simply depopulated, the signal path is broken. Replacing it with copper removes the damping.	R103.1: FPGA_CCLK [C] R103.2: FLASH_SCLK [C]
R104 [Q] 0	Zero-ohm flash signal link Depopulating it breaks the signal; a deliberate trace replacement keeps connectivity.	R104.1: FPGA_CFG_DQ0 [C] R104.2: FLASH_DQ0 [C]
R105 [Q] 33	Flash data source damping If simply depopulated, the signal path is broken. Replacing it with copper removes the damping.	R105.1: FPGA_CFG_DQ1 [C] R105.2: FLASH_DQ1 [C]
R106 [Q] 0	Zero-ohm flash signal link Depopulating it breaks the signal; a deliberate trace replacement keeps connectivity.	R106.1: FPGA_CFG_DQ2 [C] R106.2: FLASH_DQ2 [C]
R107 [Q] 0	Zero-ohm flash signal link Depopulating it breaks the signal; a deliberate trace replacement keeps connectivity.	R107.1: FPGA_CFG_DQ3 [C] R107.2: FLASH_DQ3 [C]
R108 [K] 4.7k	PROGRAM_B pull-up Configuration reset/initialization behavior can become undefined or stall.	R108.1: VCCAUX_1V8 [P] R108.2: FPGA_PROGRAM_B [C]
R109 [K] 4.7k	INIT_B pull-up Configuration reset/initialization behavior can become undefined or stall.	R109.1: VCCAUX_1V8 [P] R109.2: FPGA_INIT_B [C]
R110 [Q] 4.7k	Additional DONE pull-up DONE retains its internal pull-up; whether its rise is acceptable depends on configuration settings and loading.	R110.1: VCCAUX_1V8 [P] R110.2: FPGA_DONE [C]
R111 [S] 1k	Configuration strap M0 The deliberately fixed configuration input loses its chosen strap unless replaced by a direct connection.	R111.1: VCCAUX_1V8 [P] R111.2: CFG_M0 [C]

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Part / value	Role; consequence if removed	Actual endpoint: net [status]
R112 [S] 1k	Configuration strap M1 The deliberately fixed configuration input loses its chosen strap unless replaced by a direct connection.	R112.1: CFG_M1 [C] R112.2: GND [P]
R113 [S] 1k	Configuration strap M2 The deliberately fixed configuration input loses its chosen strap unless replaced by a direct connection.	R113.1: CFG_M2 [C] R113.2: GND [P]
R114 [S] 1k	Configuration strap PUDC_B The deliberately fixed configuration input loses its chosen strap unless replaced by a direct connection.	R114.1: VCCAUX_1V8 [P] R114.2: CFG_PUDC_B [C]
R115 [Q] 10k	JTAG TMS external idle bias External idle bias disappears; behavior then depends on device internal pulls and the actual adapter/cable.	R115.1: VCCAUX_1V8 [P] R115.2: JTAG_TMS [C]
R116 [Q] 10k	JTAG TDI external idle bias External idle bias disappears; behavior then depends on device internal pulls and the actual adapter/cable.	R116.1: VCCAUX_1V8 [P] R116.2: JTAG_TDI [C]
R117 [Q] 10k	JTAG TCK external idle bias External idle bias disappears; behavior then depends on device internal pulls and the actual adapter/cable.	R117.1: VCCAUX_1V8 [P] R117.2: JTAG_TCK [C]
R118 [Q] 33	JTAG TDO source damping If simply depopulated, the signal path is broken. Replacing it with copper removes the damping.	R118.1: FPGA_TDO [C] R118.2: JTAG_TDO [C]
R119 [Q] 33	Oscillator output damping If simply depopulated, the signal path is broken. Replacing it with copper removes the damping.	R119.1: OSC_32M_RAW [C] R119.2: CLK_32MHZ [C]
R122 [K] 15m 1% 1W	Distributed-capacitance isolation Removing it opens the rail; shorting it defeats the deliberately separated local and distributed capacitor networks.	R122.1: VAUX_REG_1V803 [P] R122.2: VCCAUX_1V8 [P]
U1 [K] XC7A100T-CSG324	Acquire, time and package ASIC data There is no FPGA function.	Complete pins: Appendix B.
U2 [K] TPS62135RGXR	Core / block ram converter This rail loses its source. The connected FPGA power pins cannot simply be left open.	Complete pins: Appendix D.
U3 [K] TPS62135RGXR	Auxiliary / configuration converter This rail loses its source. The connected FPGA power pins cannot simply be left open.	Complete pins: Appendix D.
U4 [S] TPS62135RGXR	2.5 v link bank converter This rail loses its source. The connected FPGA power pins cannot simply be left open.	Complete pins: Appendix D.
U5 [S] TPS62135RGXR	1.5 v asic-facing banks converter This rail loses its source. The connected FPGA power pins cannot simply be left open.	Complete pins: Appendix D.
U6 [S] MX25U12835FM2I-10G	Nonvolatile boot image The selected SPI self-boot path is lost; an external configuration method would be required each startup.	Complete pins: Appendix D.
Y1 [S] ASE3 32MHz 1.8V	32 MHz system reference The present application clock path stops; SPI configuration may still use its own clock, but application operation requires an alternative reference.	Complete pins: Appendix D.

B All 324 FPGA balls

The function column follows the package pin inventory; a capability in the pin name is not an implemented application assignment. U means awaiting assignment. C/P describe whole-net connectivity. G means mandatory GND correction with NC still present in the saved circuit.

Endpoint	Bank	Package function	Saved net / state	Code
U1.A1	35	IO_L9N_T1_DQS_AD7N_35	Unassigned	U
U1.A2	NA	GND	GND	P
U1.A3	35	IO_L8N_T1_AD14N_35	Unassigned	U
U1.A4	35	IO_L8P_T1_AD14P_35	Unassigned	U
U1.A5	35	IO_L3N_T0_DQS_AD5N_35	Unassigned	U
U1.A6	35	IO_L3P_T0_DQS_AD5P_35	Unassigned	U
U1.A7	35	VCCO_35	VCC_ASIC_1V5	P
U1.A8	16	IO_L12N_T1_MRCC_16	Unassigned	U
U1.A9	16	IO_L14N_T2_SRCC_16	Unassigned	U
U1.A10	16	IO_L14P_T2_SRCC_16	Unassigned	U
U1.A11	15	IO_L4N_T0_15	Unassigned	U
U1.A12	NA	GND	GND	P
U1.A13	15	IO_L9P_T1_DQS_AD3P_15	Unassigned	U
U1.A14	15	IO_L9N_T1_DQS_AD3N_15	Unassigned	U
U1.A15	15	IO_L8P_T1_AD10P_15	Unassigned	U
U1.A16	15	IO_L8N_T1_AD10N_15	Unassigned	U
U1.A17	15	VCCO_15	VCC_ASIC_1V5	P
U1.A18	15	IO_L10N_T1_AD11N_15	Unassigned	U
U1.B1	35	IO_L9P_T1_DQS_AD7P_35	Unassigned	U
U1.B2	35	IO_L10N_T1_AD15N_35	Unassigned	U
U1.B3	35	IO_L10P_T1_AD15P_35	Unassigned	U
U1.B4	35	IO_L7N_T1_AD6N_35	Unassigned	U
U1.B5	NA	GND	GND	P
U1.B6	35	IO_L2N_T0_AD12N_35	Unassigned	U
U1.B7	35	IO_L2P_T0_AD12P_35	Unassigned	U
U1.B8	16	IO_L12P_T1_MRCC_16	Unassigned	U
U1.B9	16	IO_L11N_T1_SRCC_16	Unassigned	U
U1.B10	16	VCCO_16	VCC_LINK_2V5	P
U1.B11	15	IO_L4P_T0_15	Unassigned	U
U1.B12	15	IO_L3N_T0_DQS_AD1N_15	Unassigned	U
U1.B13	15	IO_L2P_T0_AD8P_15	Unassigned	U
U1.B14	15	IO_L2N_T0_AD8N_15	Unassigned	U
U1.B15	NA	GND	GND	P
U1.B16	15	IO_L7P_T1_AD2P_15	Unassigned	U
U1.B17	15	IO_L7N_T1_AD2N_15	Unassigned	U
U1.B18	15	IO_L10P_T1_AD11P_15	Unassigned	U
U1.C1	35	IO_L16N_T2_35	Unassigned	U
U1.C2	35	IO_L16P_T2_35	Unassigned	U
U1.C3	35	VCCO_35	VCC_ASIC_1V5	P
U1.C4	35	IO_L7P_T1_AD6P_35	Unassigned	U
U1.C5	35	IO_L1N_T0_AD4N_35	Unassigned	U
U1.C6	35	IO_L1P_T0_AD4P_35	Unassigned	U
U1.C7	35	IO_L4N_T0_35	Unassigned	U
U1.C8	NA	GND	GND	P
U1.C9	16	IO_L11P_T1_SRCC_16	Unassigned	U
U1.C10	16	IO_L13N_T2_MRCC_16	Unassigned	U
U1.C11	16	IO_L13P_T2_MRCC_16	Unassigned	U
U1.C12	15	IO_L3P_T0_DQS_AD1P_15	Unassigned	U
U1.C13	15	VCCO_15	VCC_ASIC_1V5	P
U1.C14	15	IO_L1N_T0_AD0N_15	Unassigned	U
U1.C15	15	IO_L12N_T1_MRCC_15	Unassigned	U
U1.C16	15	IO_L20P_T3_A20_15	Unassigned	U
U1.C17	15	IO_L20N_T3_A19_15	Unassigned	U

Continued on next page

Endpoint	Bank	Package function	Saved net / state	Code
U1.C18	NA	GND	GND	P
U1.D1	NA	GND	GND	P
U1.D2	35	IO_L14N_T2_SRCC_35	Unassigned	U
U1.D3	35	IO_L12N_T1_MRCC_35	Unassigned	U
U1.D4	35	IO_L11N_T1_SRCC_35	Unassigned	U
U1.D5	35	IO_L11P_T1_SRCC_35	Unassigned	U
U1.D6	35	VCCO_35	VCC_ASIC_1V5	P
U1.D7	35	IO_L6N_T0_VREF_35	Unassigned	U
U1.D8	35	IO_L4P_T0_35	Unassigned	U
U1.D9	16	IO_L6N_T0_VREF_16	Unassigned	U
U1.D10	16	IO_L19N_T3_VREF_16	Unassigned	U
U1.D11	NA	GND	GND	P
U1.D12	15	IO_L6P_T0_15	Unassigned	U
U1.D13	15	IO_L6N_T0_VREF_15	Unassigned	U
U1.D14	15	IO_L1P_T0_AD0P_15	Unassigned	U
U1.D15	15	IO_L12P_T1_MRCC_15	Unassigned	U
U1.D16	15	VCCO_15	VCC_ASIC_1V5	P
U1.D17	15	IO_L16N_T2_A27_15	Unassigned	U
U1.D18	15	IO_L21N_T3_DQS_A18_15	Unassigned	U
U1.E1	35	IO_L18N_T2_35	Unassigned	U
U1.E2	35	IO_L14P_T2_SRCC_35	Unassigned	U
U1.E3	35	IO_L12P_T1_MRCC_35	Unassigned	U
U1.E4	NA	GND	GND	P
U1.E5	35	IO_L5N_T0_AD13N_35	Unassigned	U
U1.E6	35	IO_L5P_T0_AD13P_35	Unassigned	U
U1.E7	35	IO_L6P_T0_35	Unassigned	U
U1.E8	0	VCCBATT_0	GND	P
U1.E9	0	CCLK_0	FPGA_CCLK	C
U1.E10	0	TCK_0	JTAG_TCK	C
U1.E11	0	TDI_0	JTAG_TDI	C
U1.E12	0	TMS_0	JTAG_TMS	C
U1.E13	0	TDO_0	FPGA_TDO	C
U1.E14	NA	GND	GND	P
U1.E15	15	IO_L11P_T1_SRCC_15	Unassigned	U
U1.E16	15	IO_L11N_T1_SRCC_15	Unassigned	U
U1.E17	15	IO_L16P_T2_A28_15	Unassigned	U
U1.E18	15	IO_L21P_T3_DQS_15	Unassigned	U
U1.F1	35	IO_L18P_T2_35	Unassigned	U
U1.F2	35	VCCO_35	VCC_ASIC_1V5	P
U1.F3	35	IO_L13N_T2_MRCC_35	Unassigned	U
U1.F4	35	IO_L13P_T2_MRCC_35	Unassigned	U
U1.F5	35	IO_0_35	Unassigned	U
U1.F6	35	IO_L19N_T3_VREF_35	Unassigned	U
U1.F7	NA	GND	GND	P
U1.F8	NA	VCCINT	VCCINT_1V0	P
U1.F9	NA	GND	GND	P
U1.F10	NA	VCCBRAM	VCCINT_1V0	P
U1.F11	NA	GND	GND	P
U1.F12	NA	VCCAUX	VCCAUX_1V8	P
U1.F13	15	IO_L5P_T0_AD9P_15	Unassigned	U
U1.F14	15	IO_L5N_T0_AD9N_15	Unassigned	U
U1.F15	15	IO_L14P_T2_SRCC_15	Unassigned	U
U1.F16	15	IO_L14N_T2_SRCC_15	Unassigned	U
U1.F17	NA	GND	GND	P
U1.F18	15	IO_L22N_T3_A16_15	Unassigned	U
U1.G1	35	IO_L17N_T2_35	Unassigned	U
U1.G2	35	IO_L15N_T2_DQS_35	Unassigned	U
U1.G3	35	IO_L20N_T3_35	Unassigned	U
U1.G4	35	IO_L20P_T3_35	Unassigned	U
U1.G5	35	VCCO_35	VCC_ASIC_1V5	P
U1.G6	35	IO_L19P_T3_35	Unassigned	U

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Endpoint	Bank	Package function	Saved net / state	Code
U1.G7	NA	VCCINT	VCCINT_1V0	P
U1.G8	NA	GND	GND	P
U1.G9	NA	VCCINT	VCCINT_1V0	P
U1.G10	NA	GND	GND	P
U1.G11	NA	VCCBRAM	VCCINT_1V0	P
U1.G12	NA	GND	GND	P
U1.G13	15	IO_0_15	Unassigned	U
U1.G14	15	IO_L15N_T2_DQS_ADV_B_15	Unassigned	U
U1.G15	15	VCCO_15	VCC_ASIC_1V5	P
U1.G16	15	IO_L13N_T2_MRCC_15	Unassigned	U
U1.G17	15	IO_L18N_T2_A23_15	Unassigned	U
U1.G18	15	IO_L22P_T3_A17_15	Unassigned	U
U1.H1	35	IO_L17P_T2_35	Unassigned	U
U1.H2	35	IO_L15P_T2_DQS_35	Unassigned	U
U1.H3	NA	GND	GND	P
U1.H4	35	IO_L21N_T3_DQS_35	Unassigned	U
U1.H5	35	IO_L24N_T3_35	Unassigned	U
U1.H6	35	IO_L24P_T3_35	Unassigned	U
U1.H7	NA	GND	GND	P
U1.H8	NA	VCCINT	VCCINT_1V0	P
U1.H9	0	GNDADC_0	GND	P
U1.H10	0	VCCADC_0	VCCAUX_1V8	P
U1.H11	NA	GND	GND	P
U1.H12	NA	VCCAUX	VCCAUX_1V8	P
U1.H13	NA	GND	GND	P
U1.H14	15	IO_L15P_T2_DQS_15	Unassigned	U
U1.H15	15	IO_L19N_T3_A21_VREF_15	Unassigned	U
U1.H16	15	IO_L13P_T2_MRCC_15	Unassigned	U
U1.H17	15	IO_L18P_T2_A24_15	Unassigned	U
U1.H18	15	VCCO_15	VCC_ASIC_1V5	P
U1.J1	35	VCCO_35	VCC_ASIC_1V5	P
U1.J2	35	IO_L22N_T3_35	Unassigned	U
U1.J3	35	IO_L22P_T3_35	Unassigned	U
U1.J4	35	IO_L21P_T3_DQS_35	Unassigned	U
U1.J5	35	IO_25_35	Unassigned	U
U1.J6	NA	GND	GND	P
U1.J7	NA	VCCINT	VCCINT_1V0	P
U1.J8	NA	GND	GND	P
U1.J9	0	VREFN_0	GND	P
U1.J10	0	VP_0	GND	P
U1.J11	NA	VCCINT	VCCINT_1V0	P
U1.J12	NA	GND	GND	P
U1.J13	15	IO_L17N_T2_A25_15	Unassigned	U
U1.J14	15	IO_L19P_T3_A22_15	Unassigned	U
U1.J15	15	IO_L24N_T3_RS0_15	Unassigned	U
U1.J16	NA	GND	GND	P
U1.J17	15	IO_L23P_T3_FOE_B_15	Unassigned	U
U1.J18	15	IO_L23N_T3_FWE_B_15	Unassigned	U
U1.K1	35	IO_L23N_T3_35	Unassigned	U
U1.K2	35	IO_L23P_T3_35	Unassigned	U
U1.K3	34	IO_L2P_T0_34	Unassigned	U
U1.K4	34	VCCO_34	VCC_ASIC_1V5	P
U1.K5	34	IO_L5P_T0_34	Unassigned	U
U1.K6	34	IO_0_34	Unassigned	U
U1.K7	NA	GND	GND	P
U1.K8	NA	VCCINT	VCCINT_1V0	P
U1.K9	0	VN_0	GND	P
U1.K10	0	VREFP_0	GND	P
U1.K11	NA	GND	GND	P
U1.K12	NA	VCCAUX	VCCAUX_1V8	P
U1.K13	15	IO_L17P_T2_A26_15	Unassigned	U

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Endpoint	Bank	Package function	Saved net / state	Code
U1.K14	15	VCCO_15	VCC_ASIC_1V5	P
U1.K15	15	IO_L24P_T3_RS1_15	Unassigned	U
U1.K16	15	IO_25_15	Unassigned	U
U1.K17	14	IO_L1P_T0_D00_MOSI_14	FPGA_CFG_DQ0	C
U1.K18	14	IO_L1N_T0_D01_DIN_14	FPGA_CFG_DQ1	C
U1.L1	34	IO_L1P_T0_34	Unassigned	U
U1.L2	NA	GND	GND	P
U1.L3	34	IO_L2N_T0_34	Unassigned	U
U1.L4	34	IO_L5N_T0_34	Unassigned	U
U1.L5	34	IO_L6N_T0_VREF_34	Unassigned	U
U1.L6	34	IO_L6P_T0_34	Unassigned	U
U1.L7	NA	VCCINT	VCCINT_1V0	P
U1.L8	NA	GND	GND	P
U1.L9	0	DXN_0	NC in CAD	G
U1.L10	0	DXP_0	NC in CAD	G
U1.L11	NA	VCCINT	VCCINT_1V0	P
U1.L12	NA	GND	GND	P
U1.L13	14	IO_L6P_T0_FCS_B_14	FLASH_CS_B	C
U1.L14	14	IO_L2P_T0_D02_14	FPGA_CFG_DQ2	C
U1.L15	14	IO_L3P_T0_DQS_PUDC_B_14	CFG_PUDC_B	C
U1.L16	14	IO_L3N_T0_DQS_EMCCLK_14	Unassigned	U
U1.L17	14	VCCO_14	VCCAUX_1V8	P
U1.L18	14	IO_L4P_T0_D04_14	Unassigned	U
U1.M1	34	IO_L1N_T0_34	Unassigned	U
U1.M2	34	IO_L4N_T0_34	Unassigned	U
U1.M3	34	IO_L4P_T0_34	Unassigned	U
U1.M4	34	IO_L16P_T2_34	Unassigned	U
U1.M5	NA	GND	GND	P
U1.M6	34	IO_L18P_T2_34	Unassigned	U
U1.M7	NA	GND	GND	P
U1.M8	NA	VCCINT	VCCINT_1V0	P
U1.M9	NA	GND	GND	P
U1.M10	NA	VCCINT	VCCINT_1V0	P
U1.M11	NA	GND	GND	P
U1.M12	NA	VCCAUX	VCCAUX_1V8	P
U1.M13	14	IO_L6N_T0_D08_VREF_14	Unassigned	U
U1.M14	14	IO_L2N_T0_D03_14	FPGA_CFG_DQ3	C
U1.M15	NA	GND	GND	P
U1.M16	14	IO_L10P_T1_D14_14	Unassigned	U
U1.M17	14	IO_L10N_T1_D15_14	Unassigned	U
U1.M18	14	IO_L4N_T0_D05_14	Unassigned	U
U1.N1	34	IO_L3N_T0_DQS_34	Unassigned	U
U1.N2	34	IO_L3P_T0_DQS_34	Unassigned	U
U1.N3	34	VCCO_34	VCC_ASIC_1V5	P
U1.N4	34	IO_L16N_T2_34	Unassigned	U
U1.N5	34	IO_L13P_T2_MRCC_34	Unassigned	U
U1.N6	34	IO_L18N_T2_34	Unassigned	U
U1.N7	NA	VCCINT	VCCINT_1V0	P
U1.N8	NA	GND	GND	P
U1.N9	NA	VCCINT	VCCINT_1V0	P
U1.N10	NA	GND	GND	P
U1.N11	NA	VCCINT	VCCINT_1V0	P
U1.N12	NA	GND	GND	P
U1.N13	14	VCCO_14	VCCAUX_1V8	P
U1.N14	14	IO_L8P_T1_D11_14	Unassigned	U
U1.N15	14	IO_L11P_T1_SRCC_14	Unassigned	U
U1.N16	14	IO_L11N_T1_SRCC_14	Unassigned	U
U1.N17	14	IO_L9P_T1_DQS_14	Unassigned	U
U1.N18	NA	GND	GND	P
U1.P1	NA	GND	GND	P
U1.P2	34	IO_L15P_T2_DQS_34	Unassigned	U

Continued on next page

Endpoint	Bank	Package function	Saved net / state	Code
U1.P3	34	IO_L14N_T2_SRCC_34	Unassigned	U
U1.P4	34	IO_L14P_T2_SRCC_34	Unassigned	U
U1.P5	34	IO_L13N_T2_MRCC_34	Unassigned	U
U1.P6	34	VCCO_34	VCC_ASIC_1V5	P
U1.P7	0	INIT_B_0	FPGA_INIT_B	C
U1.P8	0	CFGBVS_0	GND	P
U1.P9	0	PROGRAM_B_0	FPGA_PROGRAM_B	C
U1.P10	0	DONE_0	FPGA_DONE	C
U1.P11	0	M2_0	CFG_M2	C
U1.P12	0	M0_0	CFG_M0	C
U1.P13	0	M1_0	CFG_M1	C
U1.P14	14	IO_L8N_T1_D12_14	Unassigned	U
U1.P15	14	IO_L13P_T2_MRCC_14	Unassigned	U
U1.P16	14	VCCO_14	VCCAUX_1V8	P
U1.P17	14	IO_L12P_T1_MRCC_14	CLK_32MHZ	C
U1.P18	14	IO_L9N_T1_DQS_D13_14	Unassigned	U
U1.R1	34	IO_L17P_T2_34	Unassigned	U
U1.R2	34	IO_L15N_T2_DQS_34	Unassigned	U
U1.R3	34	IO_L11P_T1_SRCC_34	Unassigned	U
U1.R4	NA	GND	GND	P
U1.R5	34	IO_L19N_T3_VREF_34	Unassigned	U
U1.R6	34	IO_L19P_T3_34	Unassigned	U
U1.R7	34	IO_L23P_T3_34	Unassigned	U
U1.R8	34	IO_L24P_T3_34	Unassigned	U
U1.R9	0	VCCO_0	VCCAUX_1V8	P
U1.R10	14	IO_25_14	Unassigned	U
U1.R11	14	IO_0_14	Unassigned	U
U1.R12	14	IO_L5P_T0_D06_14	Unassigned	U
U1.R13	14	IO_L5N_T0_D07_14	Unassigned	U
U1.R14	NA	GND	GND	P
U1.R15	14	IO_L13N_T2_MRCC_14	Unassigned	U
U1.R16	14	IO_L15P_T2_DQS_RDWR_B_14	Unassigned	U
U1.R17	14	IO_L12N_T1_MRCC_14	Unassigned	U
U1.R18	14	IO_L7P_T1_D09_14	Unassigned	U
U1.T1	34	IO_L17N_T2_34	Unassigned	U
U1.T2	34	VCCO_34	VCC_ASIC_1V5	P
U1.T3	34	IO_L11N_T1_SRCC_34	Unassigned	U
U1.T4	34	IO_L12N_T1_MRCC_34	Unassigned	U
U1.T5	34	IO_L12P_T1_MRCC_34	Unassigned	U
U1.T6	34	IO_L23N_T3_34	Unassigned	U
U1.T7	NA	GND	GND	P
U1.T8	34	IO_L24N_T3_34	Unassigned	U
U1.T9	14	IO_L24P_T3_A01_D17_14	Unassigned	U
U1.T10	14	IO_L24N_T3_A00_D16_14	Unassigned	U
U1.T11	14	IO_L19P_T3_A10_D26_14	Unassigned	U
U1.T12	14	VCCO_14	VCCAUX_1V8	P
U1.T13	14	IO_L23P_T3_A03_D19_14	Unassigned	U
U1.T14	14	IO_L14P_T2_SRCC_14	Unassigned	U
U1.T15	14	IO_L14N_T2_SRCC_14	Unassigned	U
U1.T16	14	IO_L15N_T2_DQS_DOUT_CSO_B_14	Unassigned	U
U1.T17	NA	GND	GND	P
U1.T18	14	IO_L7N_T1_D10_14	Unassigned	U
U1.U1	34	IO_L7P_T1_34	Unassigned	U
U1.U2	34	IO_L9P_T1_DQS_34	Unassigned	U
U1.U3	34	IO_L8N_T1_34	Unassigned	U
U1.U4	34	IO_L8P_T1_34	Unassigned	U
U1.U5	34	VCCO_34	VCC_ASIC_1V5	P
U1.U6	34	IO_L22N_T3_34	Unassigned	U
U1.U7	34	IO_L22P_T3_34	Unassigned	U
U1.U8	34	IO_25_34	Unassigned	U
U1.U9	34	IO_L21P_T3_DQS_34	Unassigned	U

Continued on next page

Endpoint	Bank	Package function	Saved net / state	Code
U1.U10	NA	GND	GND	P
U1.U11	14	IO_L19N_T3_A09_D25_VREF_14	Unassigned	U
U1.U12	14	IO_L20P_T3_A08_D24_14	Unassigned	U
U1.U13	14	IO_L23N_T3_A02_D18_14	Unassigned	U
U1.U14	14	IO_L22P_T3_A05_D21_14	Unassigned	U
U1.U15	14	VCCO_14	VCCAUX_1V8	P
U1.U16	14	IO_L18P_T2_A12_D28_14	Unassigned	U
U1.U17	14	IO_L17P_T2_A14_D30_14	Unassigned	U
U1.U18	14	IO_L17N_T2_A13_D29_14	Unassigned	U
U1.V1	34	IO_L7N_T1_34	Unassigned	U
U1.V2	34	IO_L9N_T1_DQS_34	Unassigned	U
U1.V3	NA	GND	GND	P
U1.V4	34	IO_L10N_T1_34	Unassigned	U
U1.V5	34	IO_L10P_T1_34	Unassigned	U
U1.V6	34	IO_L20N_T3_34	Unassigned	U
U1.V7	34	IO_L20P_T3_34	Unassigned	U
U1.V8	34	VCCO_34	VCC_ASIC_1V5	P
U1.V9	34	IO_L21N_T3_DQS_34	Unassigned	U
U1.V10	14	IO_L21P_T3_DQS_14	Unassigned	U
U1.V11	14	IO_L21N_T3_DQS_A06_D22_14	Unassigned	U
U1.V12	14	IO_L20N_T3_A07_D23_14	Unassigned	U
U1.V13	NA	GND	GND	P
U1.V14	14	IO_L22N_T3_A04_D20_14	Unassigned	U
U1.V15	14	IO_L16P_T2_CSI_B_14	Unassigned	U
U1.V16	14	IO_L16N_T2_A15_D31_14	Unassigned	U
U1.V17	14	IO_L18N_T2_A11_D27_14	Unassigned	U
U1.V18	14	VCCO_14	VCCAUX_1V8	P

C Every connector contact

J4: custom cable interface

Saved assignments only. Eight contacts remain U. SH represents four physical shell pads. The custom electrical mapping is incompatible with ordinary HDMI equipment; verify the real cable and mating carrier before use.

Endpoint	Saved net / state	Code
J4.1	VCCAUX_1V8	P
J4.2	JTAG_TMS	C
J4.3	Unassigned	U
J4.4	GND	P
J4.5	Unassigned	U
J4.6	Unassigned	U
J4.7	GND	P
J4.8	Unassigned	U
J4.9	Unassigned	U
J4.10	GND	P
J4.11	Unassigned	U
J4.12	Unassigned	U
J4.13	GND	P
J4.14	Unassigned	U
J4.15	JTAG_TDI	C
J4.16	GND	P
J4.17	JTAG_TCK	C
J4.18	JTAG_TDO	C
J4.19	LINK_12V	P
J4.SH	GND	P

J5 and J6: two 60-contact mezzanines

Every numbered contact below remains unassigned. The 117-signal proposal has not become a native contact/ball map. G is the shared ground-blade endpoint, not one of the 60 numbered signal contacts; P means the overall ground net still has gaps.

Endpoint	Saved state	Endpoint	Saved state
J5.1	Unassigned [U]	J6.1	Unassigned [U]
J5.2	Unassigned [U]	J6.2	Unassigned [U]
J5.3	Unassigned [U]	J6.3	Unassigned [U]
J5.4	Unassigned [U]	J6.4	Unassigned [U]
J5.5	Unassigned [U]	J6.5	Unassigned [U]
J5.6	Unassigned [U]	J6.6	Unassigned [U]
J5.7	Unassigned [U]	J6.7	Unassigned [U]
J5.8	Unassigned [U]	J6.8	Unassigned [U]
J5.9	Unassigned [U]	J6.9	Unassigned [U]
J5.10	Unassigned [U]	J6.10	Unassigned [U]
J5.11	Unassigned [U]	J6.11	Unassigned [U]
J5.12	Unassigned [U]	J6.12	Unassigned [U]
J5.13	Unassigned [U]	J6.13	Unassigned [U]
J5.14	Unassigned [U]	J6.14	Unassigned [U]
J5.15	Unassigned [U]	J6.15	Unassigned [U]
J5.16	Unassigned [U]	J6.16	Unassigned [U]

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Endpoint	Saved state	Endpoint	Saved state
J5.17	Unassigned [U]	J6.17	Unassigned [U]
J5.18	Unassigned [U]	J6.18	Unassigned [U]
J5.19	Unassigned [U]	J6.19	Unassigned [U]
J5.20	Unassigned [U]	J6.20	Unassigned [U]
J5.21	Unassigned [U]	J6.21	Unassigned [U]
J5.22	Unassigned [U]	J6.22	Unassigned [U]
J5.23	Unassigned [U]	J6.23	Unassigned [U]
J5.24	Unassigned [U]	J6.24	Unassigned [U]
J5.25	Unassigned [U]	J6.25	Unassigned [U]
J5.26	Unassigned [U]	J6.26	Unassigned [U]
J5.27	Unassigned [U]	J6.27	Unassigned [U]
J5.28	Unassigned [U]	J6.28	Unassigned [U]
J5.29	Unassigned [U]	J6.29	Unassigned [U]
J5.30	Unassigned [U]	J6.30	Unassigned [U]
J5.31	Unassigned [U]	J6.31	Unassigned [U]
J5.32	Unassigned [U]	J6.32	Unassigned [U]
J5.33	Unassigned [U]	J6.33	Unassigned [U]
J5.34	Unassigned [U]	J6.34	Unassigned [U]
J5.35	Unassigned [U]	J6.35	Unassigned [U]
J5.36	Unassigned [U]	J6.36	Unassigned [U]
J5.37	Unassigned [U]	J6.37	Unassigned [U]
J5.38	Unassigned [U]	J6.38	Unassigned [U]
J5.39	Unassigned [U]	J6.39	Unassigned [U]
J5.40	Unassigned [U]	J6.40	Unassigned [U]
J5.41	Unassigned [U]	J6.41	Unassigned [U]
J5.42	Unassigned [U]	J6.42	Unassigned [U]
J5.43	Unassigned [U]	J6.43	Unassigned [U]
J5.44	Unassigned [U]	J6.44	Unassigned [U]
J5.45	Unassigned [U]	J6.45	Unassigned [U]
J5.46	Unassigned [U]	J6.46	Unassigned [U]
J5.47	Unassigned [U]	J6.47	Unassigned [U]
J5.48	Unassigned [U]	J6.48	Unassigned [U]
J5.49	Unassigned [U]	J6.49	Unassigned [U]
J5.50	Unassigned [U]	J6.50	Unassigned [U]
J5.51	Unassigned [U]	J6.51	Unassigned [U]
J5.52	Unassigned [U]	J6.52	Unassigned [U]
J5.53	Unassigned [U]	J6.53	Unassigned [U]
J5.54	Unassigned [U]	J6.54	Unassigned [U]
J5.55	Unassigned [U]	J6.55	Unassigned [U]
J5.56	Unassigned [U]	J6.56	Unassigned [U]
J5.57	Unassigned [U]	J6.57	Unassigned [U]
J5.58	Unassigned [U]	J6.58	Unassigned [U]
J5.59	Unassigned [U]	J6.59	Unassigned [U]
J5.60	Unassigned [U]	J6.60	Unassigned [U]
J5.G	GND [P]	J6.G	GND [P]

D Converter, flash and oscillator pins

The actual selected-part function is shown with the saved net. In particular, U6.7 is RESET/SIO3; U4.11 is VSEL (assigned GND, physically still open). All four FB2 pins are R: NC in the saved CAD, grounding review pending. P is a net-level status and does not assert that each listed pad is isolated.

Endpoint	Selected-part function	Saved net / state	Code
U2.1	VIN	LINK_12V	P
U2.2	SW	SW_CORE	C
U2.3	GND	GND	P
U2.4	FB2	NC in CAD	R
U2.5	FB	FB_CORE	C
U2.6	VOS	VCORE_REG_1V025	C
U2.7	PG	PG_CORE	C
U2.8	EN	LINK_12V	P
U2.9	SS/TR	SS_CORE	C
U2.10	MODE	LINK_12V	P
U2.11	VSEL	GND	P
U3.1	VIN	LINK_12V	P
U3.2	SW	SW_AUX	C
U3.3	GND	GND	P
U3.4	FB2	NC in CAD	R
U3.5	FB	FB_AUX	C
U3.6	VOS	VAUX_REG_1V803	P
U3.7	PG	PG_AUX	C
U3.8	EN	PG_CORE	C
U3.9	SS/TR	SS_AUX	C
U3.10	MODE	LINK_12V	P
U3.11	VSEL	GND	P
U4.1	VIN	LINK_12V	P
U4.2	SW	SW_CFG	C
U4.3	GND	GND	P
U4.4	FB2	NC in CAD	R
U4.5	FB	FB_CFG	C
U4.6	VOS	VCC_LINK_2V5	P
U4.7	PG	PGOOD_IO	C
U4.8	EN	PG_AUX	C
U4.9	SS/TR	SS_CFG	C
U4.10	MODE	LINK_12V	P
U4.11	VSEL	GND	P
U5.1	VIN	LINK_12V	P
U5.2	SW	SW_ASIC	C
U5.3	GND	GND	P
U5.4	FB2	NC in CAD	R
U5.5	FB	FB_ASIC	C
U5.6	VOS	VCC_ASIC_1V5	P
U5.7	PG	PGOOD_IO	C
U5.8	EN	PG_AUX	C
U5.9	SS/TR	SS_ASIC	C
U5.10	MODE	LINK_12V	P
U5.11	VSEL	GND	P
U6.1	CS#	FLASH_CS_B	C
U6.2	SO / SIO1	FLASH_DQ1	C
U6.3	WP# / SIO2	FLASH_DQ2	C
U6.4	GND	GND	P
U6.5	SI / SIO0	FLASH_DQ0	C
U6.6	SCLK	FLASH_SCLK	C
U6.7	RESET# / SIO3	FLASH_DQ3	C
U6.8	VCC	VCCAUX_1V8	P
Y1.1	Enable	VCCAUX_1V8	P
Y1.2	GND	GND	P
Y1.3	Clock output	OSC_32M_RAW	C
Y1.4	VDD	VCCAUX_1V8	P

End of reference. All 130 component references and all 762 distinct numbered endpoints are included. The visual maps and text examples repeat selected identifiers for explanation; the primary appendix endpoint rows contain each endpoint once.